

SOFTWARE DEVELOPMENT PORTFOLIO

Mohamad Hasan

Full-Stack Developer — Flutter & Laravel

I design and ship production mobile apps and backend platforms end to end. From offline-first Flutter apps with device licensing, thermal printing and push notifications, to Laravel ERPs and multi-vendor marketplaces with clean service layers, role-based access and versioned REST APIs. Plus a hardware side — Arduino and ESP32 control systems running in the real world.

12

PROJECTS

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FLUTTER APPS

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LARAVEL BACKENDS

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EMBEDDED SYSTEMS

PART ONE

Flutter Mobile Apps

Five cross-platform mobile applications — offline-first architecture, local SQLite/Drift databases, Bluetooth thermal printing, barcode scanning, Firebase push, maps and payments, and device-bound subscription licensing. Arabic-first (RTL) and multi-market.

01	EcoPack	Flutter
02	Smart Agent	Flutter
03	Fawateer	Flutter
04	Accounting Book	Flutter
05	Al Ghadeer	Flutter

FLUTTER

EcoPack

A smart waste-cleaning service — schedule and pay for recycling pickups

A customer-facing app for scheduling and paying for waste-collection and recycling pickups. Residents register, save home addresses with map-based location picking, browse subscription plans, and place recurring pickup orders (bag counts, days, time slots). Localized for the Russian market, with Stripe payments and Firebase push.

KEY FEATURES

- Token auth, registration and profile with photo upload
- Address book with a 2GIS map picker and geolocation
- Browse subscription offers and discounted plans
- Recurring pickup orders with history and tracking
- Stripe payments for orders
- FCM push, in-app inbox, unread badges, mark-all-read
- Promo banner carousel and light/dark/system theming

TECH STACK

Flutter 3.8

Provider

REST API (http)

flutter_stripe

2GIS Maps SDK

geolocator

Firebase Messaging

local notifications

secure_storage

image_picker

ENGINEERING HIGHLIGHTS

Full FCM lifecycle (foreground, background and terminated) with token sync back to the backend. 2GIS Maps integration behind a custom location-picker widget with a dynamically fetched API key. Stripe checkout, and English/Russian localization scaffolding.

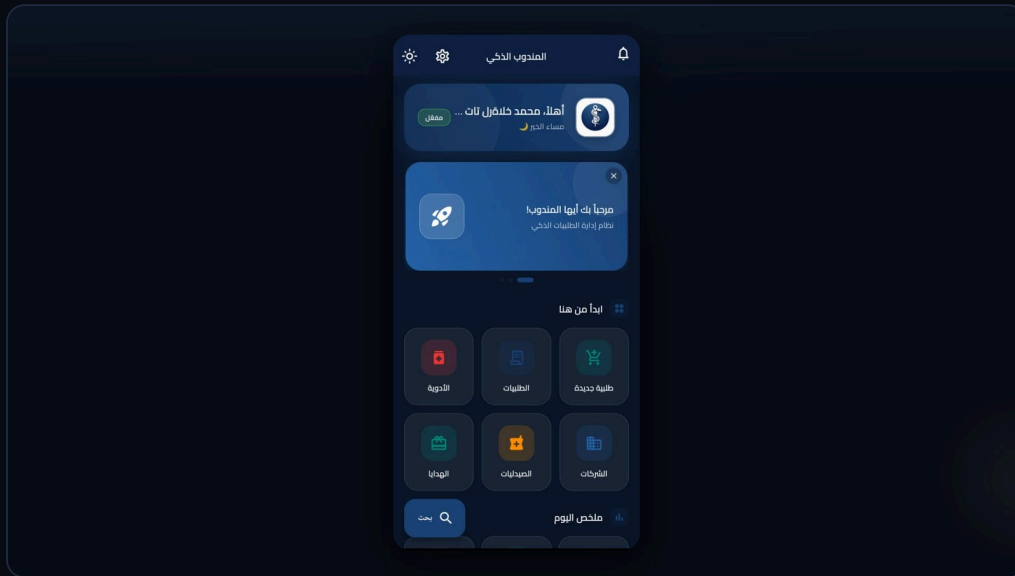
Domain: **Waste / recycling**Market: **Russia**Backend: **Laravel REST**Payments: **Stripe**

FLUTTER

Smart Agent المندوب الذكي

Offline-first order and invoicing app for medicine sales representatives

A fully offline order-management tool for pharmaceutical sales reps. Reps maintain a local catalog of companies, medicines (USD/SYP pricing), pharmacies, warehouses and gifts, then create orders and invoices exported to PDF or printed on Bluetooth thermal printers. Arabic-first (RTL), targeting the Syrian market, on a licensing model with a trial period.



KEY FEATURES

- Offline CRUD catalogs: companies, medicines, pharmacies, warehouses, gifts
- Orders with line items and optional gift quantities
- PDF invoices (Cairo Arabic font) and filtered PDF reports
- Bluetooth thermal printing with QR codes
- Device activation and subscription licensing with trial mode
- Google Drive backup/restore, plus import, export and share
- Global search and FCM notification history

TECH STACK

Flutter 3.x

Provider + get_it

sqlite (v14 schema)

pdf / printing

blue_thermal_printer

qr_flutter

googleapis (Drive)

Firebase Messaging

device_info + crypto

ENGINEERING HIGHLIGHTS

Feature-first architecture (core/ + features/) behind a DI service locator. Licensing split into single-responsibility services — ActivationService, DeviceIdentityService with salted SHA-256, TimeTamperGuard, OfflineLimitGuard and TrialModeService. Versioned SQLite migrations, and full RTL Arabic with custom slide routes.

Domain: Pharma sales

Mode: Offline-first

Database: SQLite v14

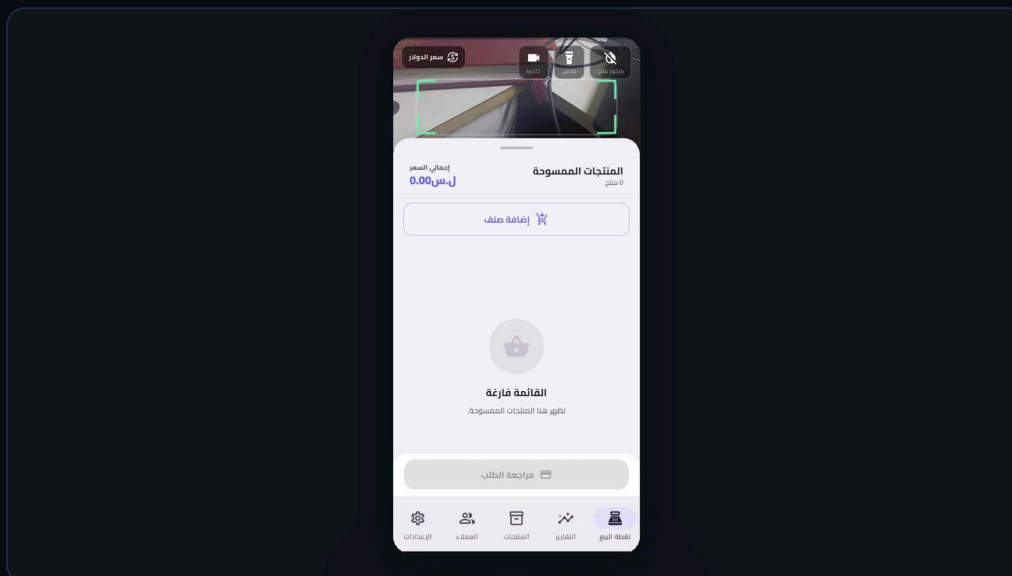
Printing: Bluetooth ESC/POS

FLUTTER

Fawateer فواتير

Offline-first Arabic POS, billing and invoicing for small shops

An offline-first point-of-sale app for small retailers — barcode scanning, Bluetooth thermal receipt printing, product and inventory management, and sales invoices with history. It also handles credit sales with a customer debt ledger, and gates commercial use behind subscription licensing. Arabic-first (RTL) with English as a secondary locale.



KEY FEATURES

- Live continuous barcode scanning in the POS screen
- Bluetooth thermal receipts — Arabic rendered as raster bitmaps
- Product and inventory CRUD with cost and low-stock alerts
- Sell-by-weight with live weight ↔ money entry
- Sales invoices with full stream-backed history
- Customer debt ledger and shareable account statements
- Subscription gate with a 72h offline grace period and FCM live-unlock

TECH STACK

Flutter 3.x

BLoC + get_it

Clean Architecture

Drift (SQLite v8)

go_router

fpdart (Either)

mobile_scanner

print_bluetooth_thermal

Firebase Messaging

ENGINEERING HIGHLIGHTS

Clean Architecture per feature (domain / data / presentation) with a typed Failure taxonomy mapped to localized strings. Disciplined append-only Drift migrations (v8) with partial-unique indexes. Device-identity licensing via SHA-256, and a GoRouter StatefulShell with five tabs behind a licensing redirect gate.

Domain: **Retail POS**Architecture: **Clean + BLoC**Database: **Drift v8**Scanning: **mobile_scanner**

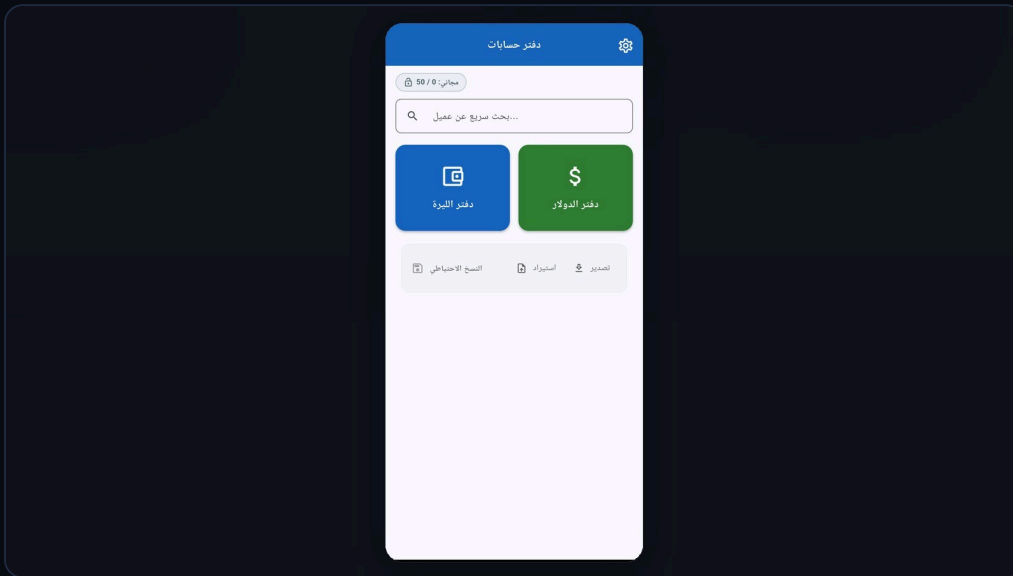
FLUTTER

Accounting Book

دفتر حسابات

Offline multi-currency ledger for tracking customer debts and credits

A personal and small-business accounting app that lets a shop owner record who owes them money and whom they owe, organised by customer and group. It works entirely offline on local SQLite, supports multi-currency (local vs. dollar) balances, and generates, prints and shares PDF account statements. Distribution is gated behind device-bound activation.



KEY FEATURES

- Customer management with grouping and transaction history
- Debit/credit transactions on a due-vs-paid convention
- Multi-currency accounts with UI filtering by currency
- PDF account statements for print or share
- Import/export with daily background auto-backup
- App lock: PIN with brute-force lockout, plus biometric unlock
- Device activation and in-app update checking

TECH STACK

Flutter 3.10 · M3

Provider

sqflite

pdf / printing

secure_storage

local_auth

workmanager

sentry_flutter

crypto (SHA-256)

ENGINEERING HIGHLIGHTS

Robust startup with runZonedGuarded and a crash service, plus deferred background init for faster launch. Two-layer crash reporting — a local capped log and DSN-gated Sentry. Centralized design-system tokens and shared widgets. PIN stored as salted SHA-256 in encrypted storage, and Arabic-first RTL with capped text scaling.

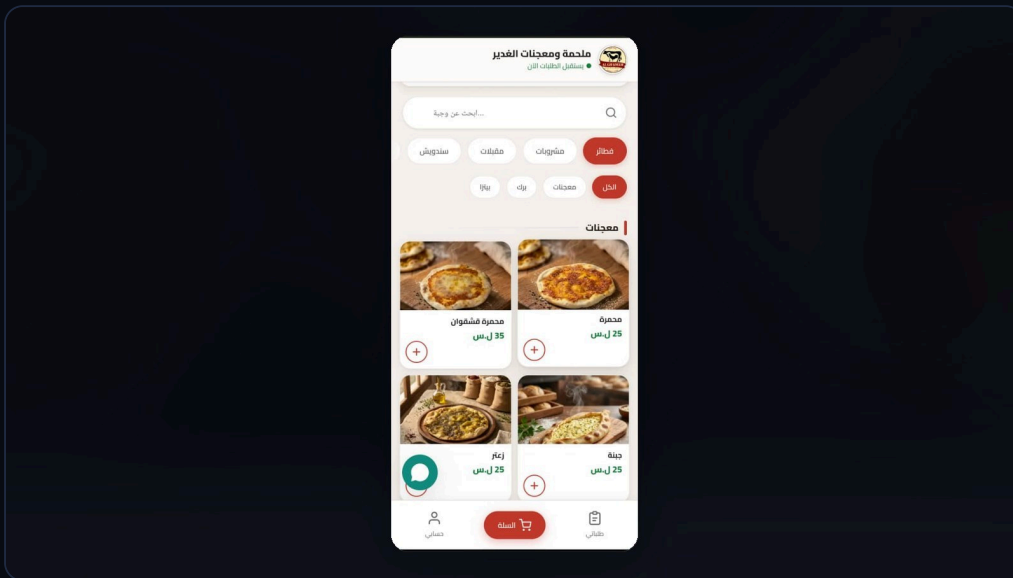
Domain: **Accounting**Mode: **Fully offline**Database: **SQLite**Security: **PIN + biometric**

FLUTTER

Al Ghadeer ملحة ومعجنات الغدير

Native Android shell for the Al Ghadeer butchery and pastries ordering site

A Flutter WebView wrapper that packages the Al Ghadeer restaurant website (Laravel + React) into a native Android app, layering native capabilities on top of the web experience: push notifications, offline detection, a native splash, and system routing for phone, WhatsApp and email links. Pairs with the Laravel ERP shown below.



KEY FEATURES

- Full-screen WebView with a native splash and logo screen
- FCM push with foreground and background auto-display
- Deep links — a notification URL opens directly in the WebView
- Offline error screen with auto-retry on reconnect
- Smart back navigation and press-again-to-exit
- tel, mailto, WhatsApp and off-domain links routed to the system
- Auto-granted WebView permissions and native Arabic dialogs

TECH STACK

Flutter 3.10

webview_flutter

Firebase Messaging

local notifications

connectivity_plus

url_launcher

native_splash

ENGINEERING HIGHLIGHTS

A native ↔ web token bridge: a strict JS-channel contract shares the FCM token with the deployed web app. Hardened token acquisition with 5× exponential backoff plus callback polling, a custom User-Agent marker so the React site can detect the native shell, and a documented end-to-end backend ↔ web ↔ Flutter ↔ FCM flow.

Domain: Food retail

Type: WebView shell

Push: FCM v1

Bridge: JS channel

PART TWO

Laravel Backends & Platforms

Four backend platforms built on Laravel 12 / PHP 8.2 — clean service layers, policy-based authorization, versioned REST APIs powering Flutter apps and React SPAs, financial and accounting modules, and a large-scale multi-vendor marketplace built with DDD.

06	Al Ghadeer Restaurant ERP	Laravel 12
07	Medical Inventory ERP	Laravel 12
08	Trydos Market	Laravel 12
09	EvoTech Core	Flagship platform

LARAVEL 12

AI Ghadeer Restaurant ERP

Bilingual restaurant ERP — a Blade admin panel and a React ordering SPA on one API

A restaurant management backend powering two front-ends from a single codebase: a Blade + Bootstrap/Tailwind admin panel for staff, and a React 18 customer SPA for online ordering. It handles the full order lifecycle (table, delivery, takeaway), menu and catalog, delivery zones, weight-based and customizable products, payments, reporting and Firebase push. It also backs the AI Ghadeer mobile app.



KEY FEATURES

- Forward-only order workflow: pending → accepted → ready → delivered
- Three order types with scheduled delivery and per-zone fees
- Two-level menu with weight-based and customizable products
- Payment tracking that blocks completion until paid
- Customer SPA ordering with a localStorage cart and token auth
- Reports and analytics — revenue, order type/status, top products
- Bulk Excel product import/export; bilingual AR/EN with RTL

TECH STACK

Laravel 12 · PHP 8.2

Service layer

MySQL

API Resources

React 18 SPA (Vite)

Blade + Bootstrap/Tailwind

kreait/Firebase FCM

PhpSpreadsheet

ENGINEERING HIGHLIGHTS

Audience-split controllers (Admin / API V1 / Public) sharing an ApiResponse trait. Session isolation via a separate customer-SPA cookie, so customer traffic can't corrupt admin auth. Roles and a ProductPolicy gate. FCM through a google/auth service account dispatched as a queued job. A versioned product cache with model-event busting, and an /api/health DB probe.

Domain: Restaurant

Clients: SPA + Flutter + admin

API: REST V1

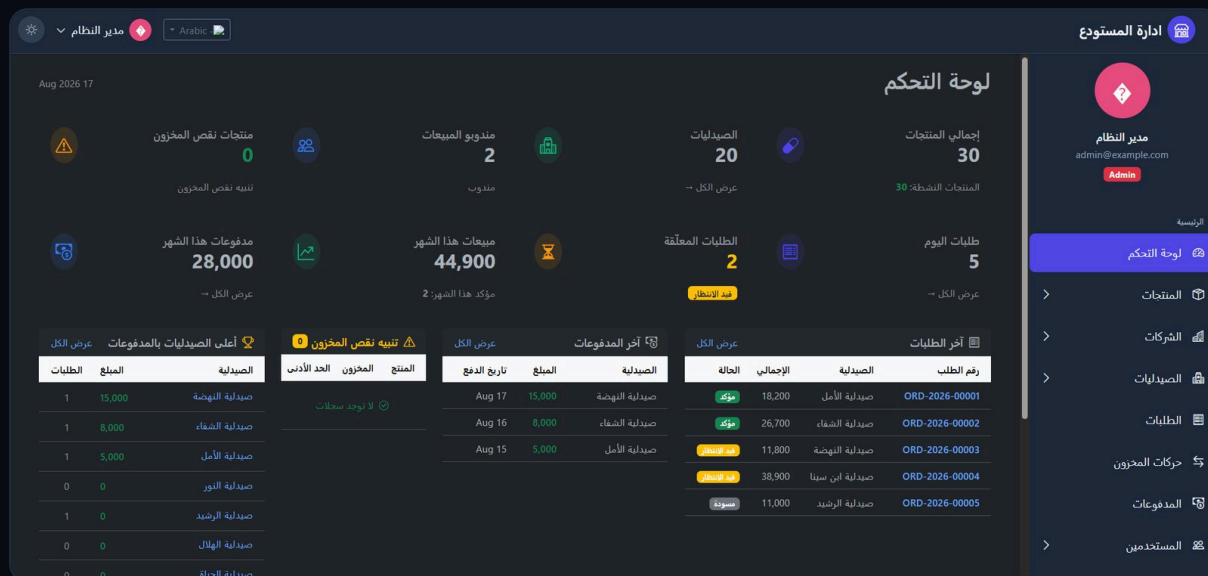
Push: FCM

LARAVEL 12

Medical Inventory ERP

Pharmaceutical distribution ERP — stock, sales orders, payments and accounting

A Laravel backend managing a medical and pharmaceutical distribution business, connecting a central company and warehouse with pharmacies through field sales reps. It tracks product stock and pricing, sales orders placed at pharmacies, payments and collections, and per-pharmacy financial account statements. It exposes a versioned REST API with offline-first sync for a Flutter app, plus a Blade admin panel.



KEY FEATURES

- Product catalog with per-product pricing and stock levels
- Sales order lifecycle: create, confirm and cancel with line items
- Stock-movement tracking through a dedicated StockService
- Payments and collections recorded per pharmacy
- Per-pharmacy financial statements and ledger
- Rep and admin dashboards with aggregates
- Offline-first mobile sync — bootstrap plus incremental changes

TECH STACK

Laravel 12 · PHP 8.2

Service + Policies

MySQL

Passport OAuth2

REST /api/v1

spatie/activitylog

PhpSpreadsheet

ENGINEERING HIGHLIGHTS

Clean layered architecture (controllers → services → models) with authorization policies. Role-based scoping, so reps see only their own pharmacies and orders while admins keep broad scope. A company/pharmacy/rep relational model approximating light multi-tenancy, Spatie audit logging, and an accounting layer (account entries and statements) that goes beyond plain inventory.

Domain: Pharma distribution

Auth: Passport OAuth2

Sync: Offline-first

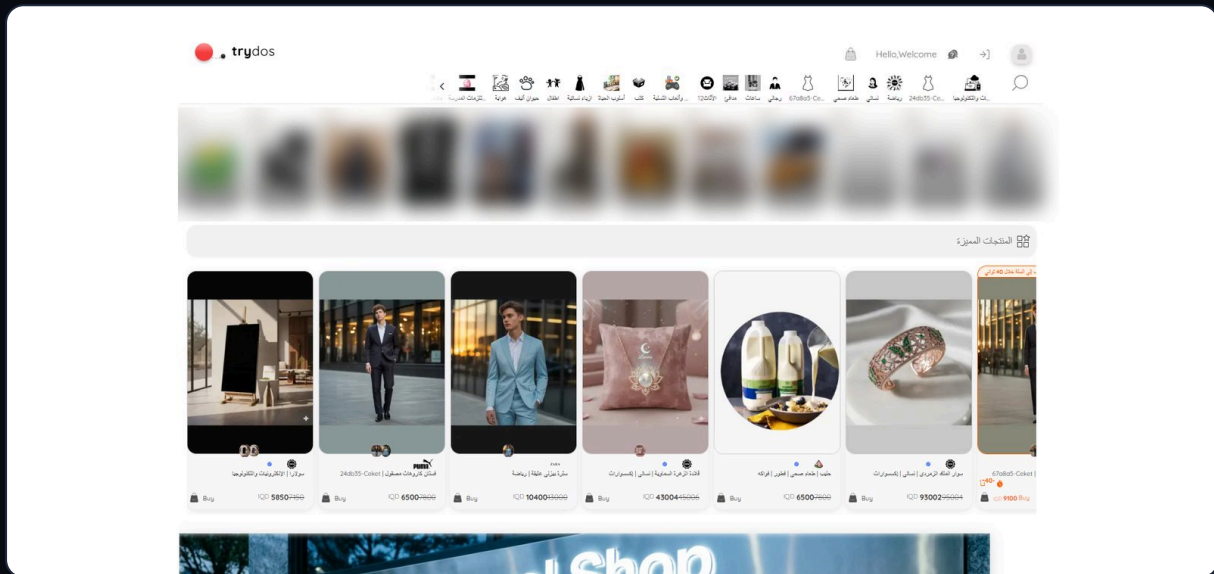
API: REST V1

LARAVEL 12

Trydos Market

Large-scale multi-vendor e-commerce platform with POS, wallet and logistics

A comprehensive Laravel backend powering a multi-vendor online marketplace serving customers, sellers and administrators. It handles product catalog and inventory, cart and checkout, order processing and returns/refunds, seller shops and boutiques, wallet and loyalty points, and integrations with payment gateways, shipping carriers, search and messaging — built on a Domain-Driven Design.



KEY FEATURES

- Multi-vendor shops, boutiques and seller analytics
- Full catalog: categories, brands, attributes, flash deals, campaigns
- Cart, coupons, wishlists, reviews and abandoned-cart recovery
- Orders, invoices and a return/refund request workflow
- Wallet and loyalty points with withdrawal requests
- POS and delivery-man (last-mile) API endpoints
- Seller ↔ customer chat, tickets and multi-channel notifications

TECH STACK

Laravel 12 · PHP 8.2

Domain-Driven Design

Passport + Sanctum + Socialite

Redis

Elasticsearch

Octane + Horizon

Stripe / Razorpay / MercadoPago / Telr

Odoo ERP sync

Gemini translation

ENGINEERING HIGHLIGHTS

Extensive DDD bounded contexts — Orders, Products, Wallet, Returns, Seller, Invoice and more. Granular RBAC including per-seller shop roles with translated names. Multi-gateway payments plus external shipping (SkyNet), Odoo ERP product and order sync, Elasticsearch search, and heavy scheduling and background jobs with Telegram API and comprehensive audit logging.

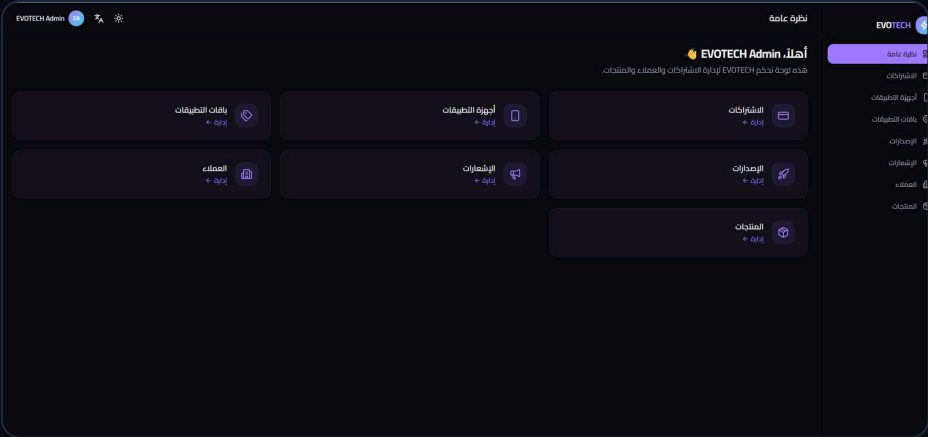
Domain: **E-commerce**Architecture: **DDD**Scale: **Octane + Redis + ES**Gateways: **5+**

FLAGSHIP PLATFORM

EvoTech Core

The control plane and licensing/billing backbone for an entire product ecosystem

An API-first platform that every EvoTech product — Smart Agent, Fawateer, Accounting Book, the restaurant ERP/POS suite, the pharma warehouse, IoT controllers — authenticates against, is licensed by, billed through and reports to, via one central REST API. Three surfaces in one: a bilingual marketing website, an authenticated subscriptions/licenses/billing dashboard, and a product-to-platform integration API. A Laravel 12 modular monolith built to scale to 100+ products and 1000+ tenants.



KEY FEATURES

- Sanctum auth with a password policy and per-account/IP throttling
- Multi-tenant companies with auto-scoped customers
- Subscription lifecycle with plan snapshotting and an expiry sweep
- Licensing: activation slots, suspend/revoke, and an event ledger
- Ed25519-signed offline license tokens for devices and IoT
- Billing: auto-issued invoices and an immutable payment ledger
- Download Center with checksummed releases and product self-update
- Notifications, audit log and KPI reports across modules

TECH STACK

Laravel 12 · PHP 8.4

Sanctum 4

MySQL / SQLite (test)

Scramble OpenAPI 3.1

Ed25519 / ext-sodium

Larastan (max) · Pint · CI

Next.js 16 · React 19

TypeScript strict

Tailwind v4 · shadcn/ui

TanStack Query

next-intl · RTL

ENGINEERING HIGHLIGHTS

A modular monolith with microservice-ready seams — 14 auto-discovered domain modules communicating only via contracts and domain events, with no cross-module joins, governed by a formal architecture constitution and ADRs. Strict four-layer modules (presentation → application → domain → infrastructure) with DTO boundaries. URI-versioned /api/v1 with standard {data, meta, links} success envelopes and machine-readable {error:{code, message, trace_id}} failures. Three auth audiences (staff Sanctum, product API-key guard, public signed URLs), UUIDv7 route keys over internal PKs, and event-driven side effects (SubscriptionActivated → auto-license + auto-invoice). Arabic-RTL-default bilingual on both tiers, with decoupled deployment across website / api / app subdomains via PM2 and Nginx.

Modules: **14 domains** Auth: **3 audiences** Scale target: **100+ products** Frontend: **Next.js 16**

PART THREE

Embedded & Hardware

Three microcontroller systems that run in the physical world — a call-to-prayer controller installed in a mosque, a Wi-Fi home automation node, and a multi-rotor UAV built from the frame up. Where the engineering degree meets the software career.

10	Automatic Call-to-Prayer System	Embedded
11	Smart Home Control System	Embedded
12	Unmanned Aerial Vehicle (Drone)	Graduation project

EMBEDDED

Automatic Call-to-Prayer System

A circuit that broadcasts the adhan on time, every day, with no one touching it

A microcontroller circuit designed and built to broadcast the call to prayer automatically at the correct time each day, with no manual intervention. It is installed and running in a mosque.



KEY FEATURES

- Real-time clock (RTC) module with stored daily prayer timetables
- Relay switching that drives the mosque amplifier directly
- LCD interface for time display and settings
- Manual override controls for the mosque staff
- Runs unattended — no phone, no internet, no PC required

TECH STACK

Arduino

C / C++

RTC module (DS3231)

Relay switching

LCD display

Push-button UI

ENGINEERING HIGHLIGHTS

The design goal was zero maintenance: the timetable lives in the device, the RTC keeps time across power cuts, and the relay stage isolates the low-voltage logic from the amplifier line. Manual override was added so staff are never locked out of their own sound system.

Type: **Embedded control**Status: **Deployed in a mosque**Clock: **RTC module**Output: **Relay → amplifier**

EMBEDDED

Smart Home Control System

Switch the lights from anywhere — over a Telegram bot or a mobile app

An ESP32-based Wi-Fi controller for household lighting and appliances, operated remotely from anywhere with an internet connection. Control runs through two independent front-ends: a Telegram bot and a mobile application.

KEY FEATURES

- ESP32 Wi-Fi controller for lighting and household appliances
- Control through a Telegram bot — no app install needed
- Control through a dedicated mobile application
- Relay modules switching the mains circuits
- Device status feedback so the user can check each output remotely

TECH STACK

ESP32

C / C++

Wi-Fi

Telegram Bot API

Mobile app

Relay modules

ENGINEERING HIGHLIGHTS

Two control paths were deliberate: the Telegram bot means any family member can operate the house from a chat they already have open, while the mobile app gives a proper dashboard. Status feedback closes the loop — you always know whether the command actually landed.

Type: **IoT controller**Board: **ESP32**Control: **Telegram + app**Output: **Relay modules**

GRADUATION PROJECT

Unmanned Aerial Vehicle (Drone)

Built from frame to flight — mechanics, electronics, control software and test flights

My graduation project for the Computer Systems Engineering degree: a multi-rotor unmanned aerial vehicle designed and built from scratch, covering the mechanical build, the power and motor electronics, the stabilisation control loop, and flight testing.

KEY FEATURES

- Multi-rotor airframe assembly and balancing
- Stabilisation using gyroscope and accelerometer data in a microcontroller control loop
- Brushless motor and ESC control
- Power distribution and battery management
- Wireless remote-control communication link
- Full cycle: mechanical assembly, electronics, control software, calibration and flight testing

TECH STACK

Microcontroller

C / C++

Gyroscope + accelerometer

PID control loop

Brushless motors + ESC

RC link

Power distribution

ENGINEERING HIGHLIGHTS

The hard part was never the hardware — it was the control loop. Reading the gyroscope and accelerometer is easy; fusing them into a stable attitude estimate and tuning the response so the airframe holds level instead of oscillating is where the project actually lived.

Type: UAV / multi-rotor

Year: 2020

Institution: University of Aleppo

Scope: Full build

GET IN TOUCH

Let's build something.

Thank you for reviewing my work. I'm available for Flutter mobile development, Laravel backend engineering and full-stack product delivery.

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